

The Diluted Signal: GPA Inflation and Entry-Level Wages

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Research Question

Does GPA affect starting wages and how strong is it as a labor market signal?

Does controlling for background factors change the relationship?

Why this matters:

- Students invest heavily in GPA
- Employers use GPA in hiring decisions
- Important to understand its economic value



Economic Intuition

GPA may increase wages because:

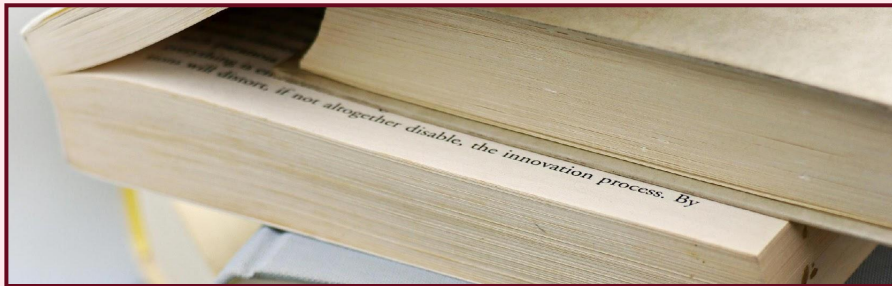
- Signals ability/productivity

- Reflects work ethic

But:

- Employers may rely more on experience

- GPA may be less important after first

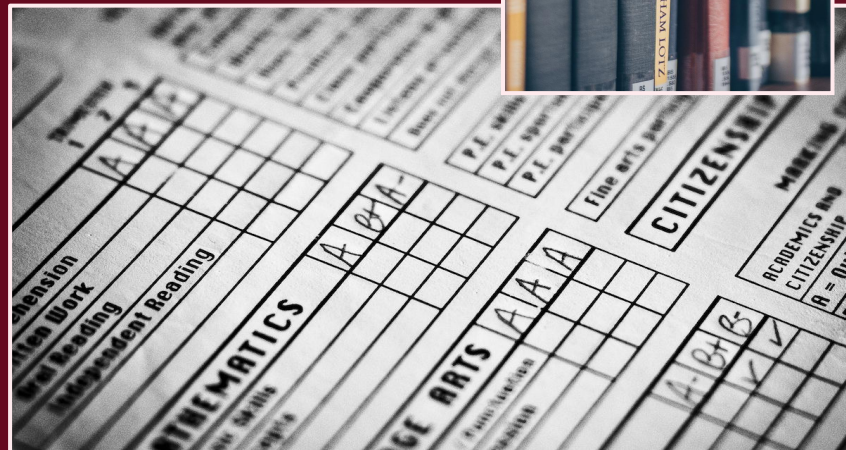


Hypothesis:

Higher GPA $\rightarrow$ higher wages

Regression sample: N = 516

- GPA (0–4 scale)
- First job wage (Job1_Wage)
- Parent education
- Graduation year (Degree_Date)
 - Timeline: Graduation dates



Summary Statistics

	count	mean	std	min	max
GPA	1380.0	2.604957	0.846797	0.14	4.0
Parent_Highest_Grade	1380.0	14.249275	2.929959	2.0	20.0
Job1_Wage	516.0	38430.055814	25577.787418	10400.0	233334.4
HH_Income_at_17	1063.0	57377.432738	44758.961944	0.0	246474.0
Degree_Date	531.0	2005.348399	2.467351	1999.0	2013.0

Notes:

Average GPA: 2.60

Average starting wage: \$38,430

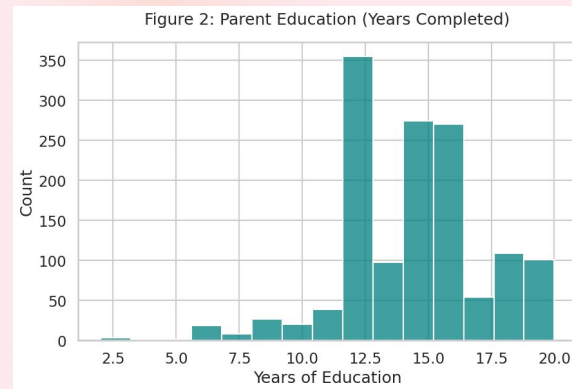
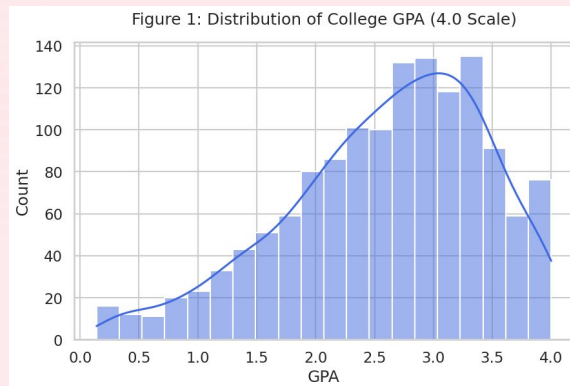
Parent education: 14 years (avg)

High variation in wages (std $\approx$ \$25,500)

Interpretation:

GPA shows meaningful variation

Wages are highly dispersed



Model 1 : GPA only

$$\text{Wage} = \beta_0 + \beta_1 \cdot \text{GPA}$$

Results

GPA coefficient: 5,316

Standard error: 2,328

p-value: 0.023

R²: 0.010

Interpretation

+\$5,316 increase in wages per 1 GPA point

Model 2 : Adding Controls

$$\text{Wage} = \beta_0 + \beta_1 \cdot \text{GPA} + \beta_2 \cdot \text{Parent Education}$$

Results

GPA: 4,743 ($p = 0.045$)

Parent Education: 509 (not significant)

R²: 0.014

Key Insight

GPA effect decreases slightly

Model Comparison

Model	GPA Effect	Significant	R ²
Model 1	5,316	Yes	0.010
Model 2	4,743	Yes	0.014

Conclusion:

- GPA effect is robust
- Very low explanatory power

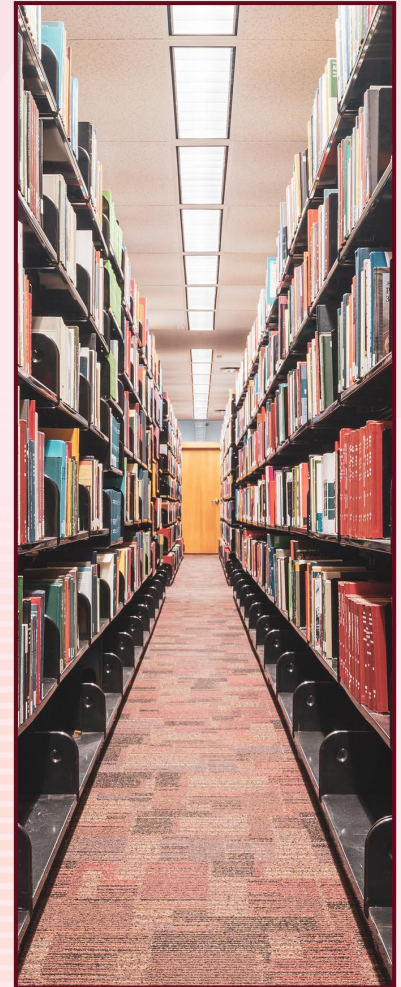
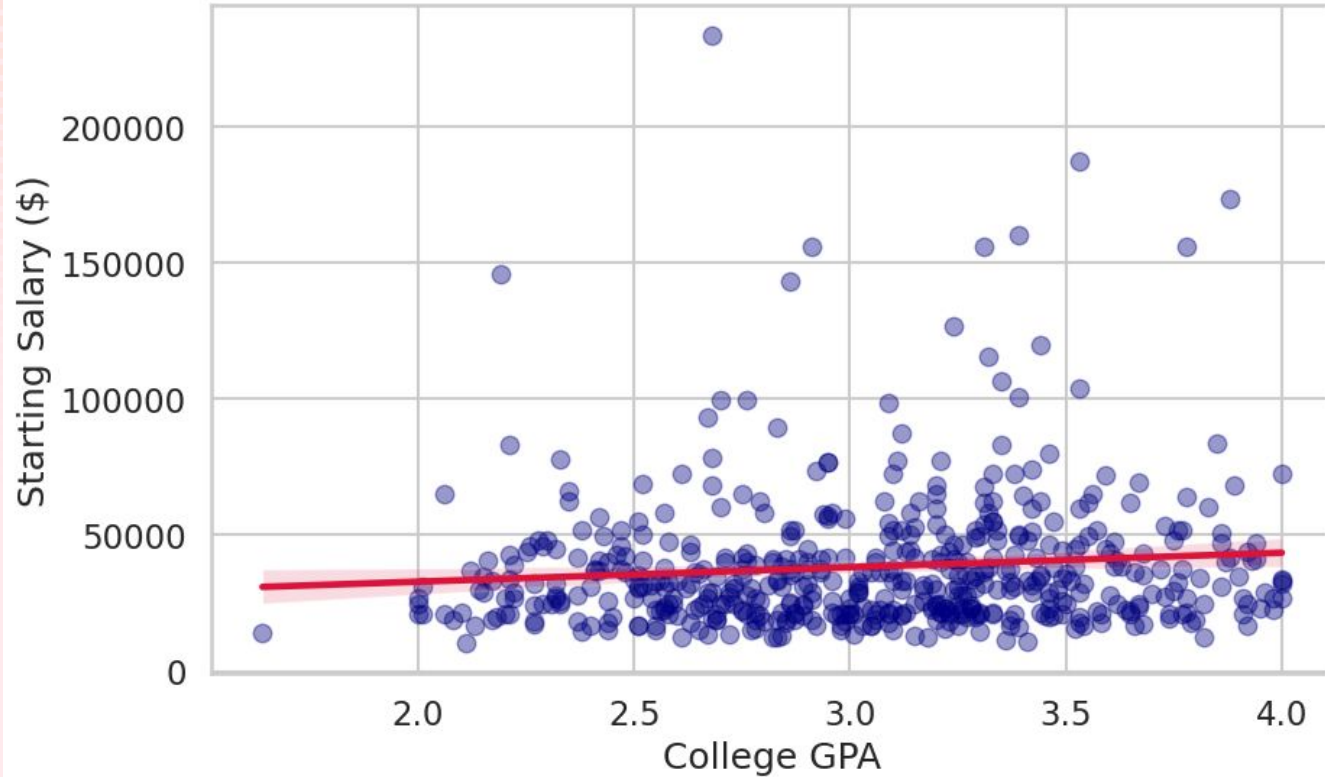


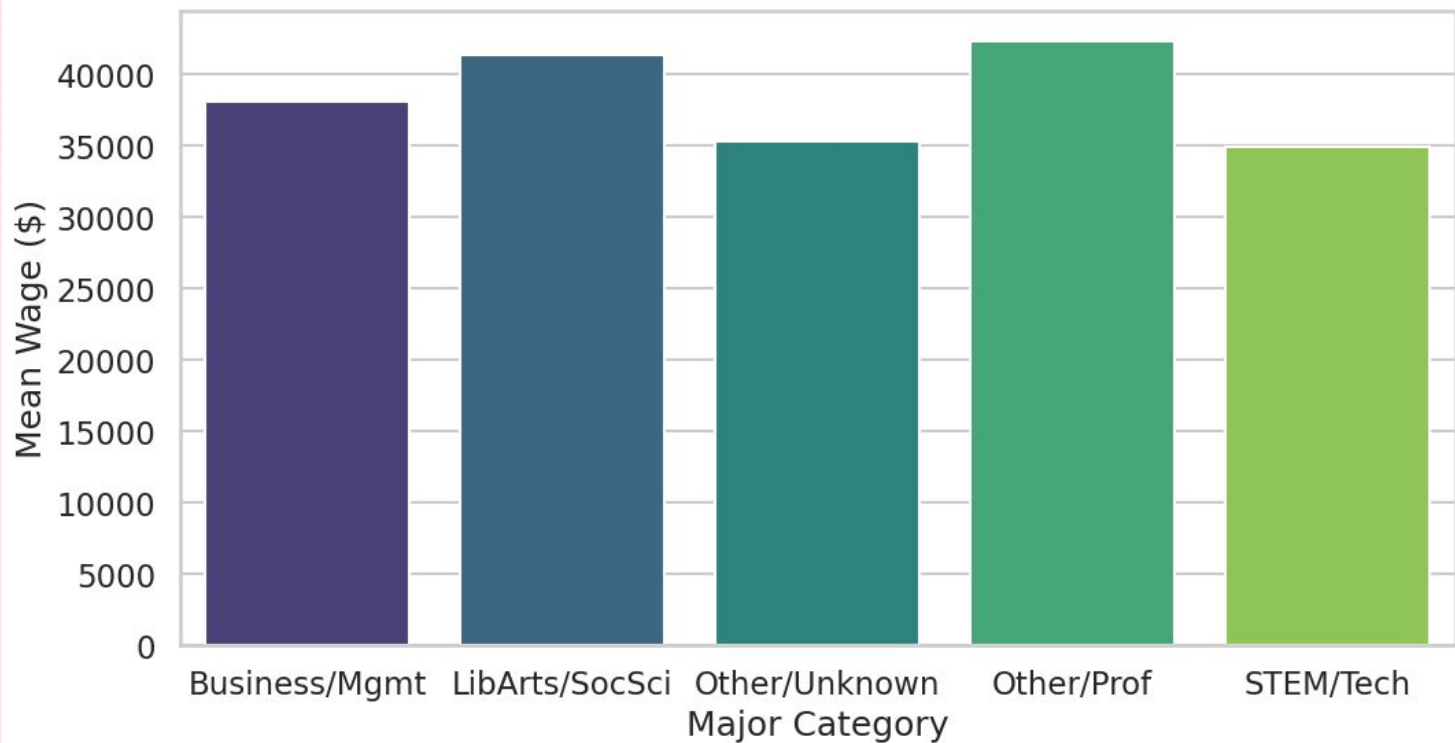
Figure 3: The Academic Signal (GPA vs. Job 1 Wage)



GPA vs Starting Wage

- ★ Positive trend
- ★ High dispersion

Figure 5: Average Starting Salary by Major Group



Wages by Major Group

- ★ Differences across majors
- ★ Bigger variation than GPA effect

Model 3 : Expanded Model

Figure 7: Mean Wage Growth (First 4 Jobs)

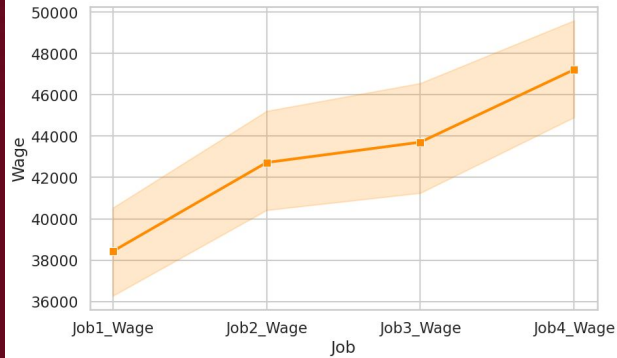
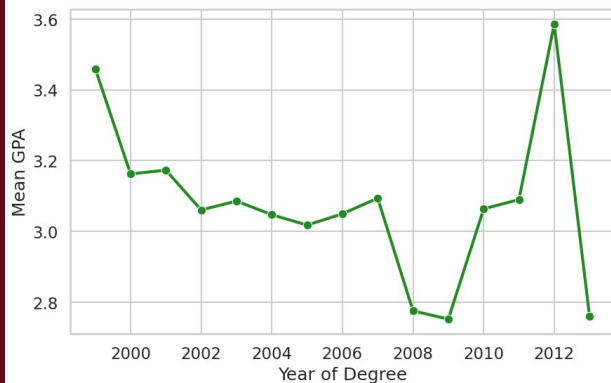


Figure 4: Average GPA by Graduation Year (1998-2013)



Includes:

- GPA
- Parent education
- Graduation year
- GPA × Year interaction
- Major fixed effects

Results

GPA not significant

Interaction not significant

$R^2 = 0.095$

Model suffers from multicollinearity

Is GPA Losing Value?

- Tested using GPA $\times$ Degree_Date
- Coefficient: negative but not significant

No strong evidence that GPA's value is declining over time

While there are hints of GPA inflation, the results are not statistically strong.



Limitations

- Missing wage data (selection bias)
- No control for ability or school quality
- High wage variation and outliers
- Results are correlational, not causal

Conclusion

GPA matters, but much less than expected

- GPA has a positive effect on wages
- Effect is statistically significant but small
- Explains very little of wage variation

Implications:

- Students: GPA matters less than expected
- Employers: GPA may be imperfect signal
- Future research: Add experience & Better wage timing



Thank You.
